

Ava Farkash

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Education

Cornell University (Ithaca, NY)

Expected May 2027

Bachelors of Science - Mechanical Engineering

Relevant Courses: Dynamics, Introduction to Mechanical Design, Introductory Fluid Mechanics, Mechanics of Engineering Materials, System Dynamics, Heat Transfer, Mechatronics

Technical Projects

- Testing Rig for Landing Gear - Design Build Fly *October 2024*
 - Designed and built a wooden test rig for the nose gear to estimate the maximum applied load in tension the gear could tolerate before failure; data collection by digital hanging scale and measured displacement
- Carbon Fiber Fuselage Fabrication - Design Build Fly *January 2025 and 2026*
 - Learned and applied an epoxy based carbon fiber layup method with custom 3D printed molds to produce a custom fuselage
- Passenger Compartment Creation - Design Build Fly *September 2025-February 2026*
 - Designed/CADed, prototyped, and constructed (3D printed ABS) a passenger compartment to ensure security within the fuselage
- Fluid Mechanical Dissection - Introductory Fluid Mechanics *November-December 2025*
 - Documented the process of disassembling a gas powered air blower to explore the fluid mechanical systems inside the machine; created a short video describing functionality, the fluid concepts employed, and used actual dimensions from the blower to calculate the Reynold's Number and airflow velocity at different sections

Professional Experience

Mechanical Sub-team Member

October 2024-Present

- Responsible for designing and manufacturing the fuselage, landing gear, and passenger/cargo requirements; work in tandem with aerodynamics and propulsion sub-teams to integrate their components; participated in the 2026 AIAA DBF competition in Wichita, Kansas

Arc Building Partners - Field Operations Intern

June - August 2025

- Designed and implemented a comprehensive hard drive system to archive all closeout documentation for the Albright Knox Gundlach Art Museum project; presented its structure and usage guidelines to the Director of Facilities, Planning, and Management
- Shadowed project engineers, head safety officers, and project superintendents at various sites

Performing and Media Arts Department - Special Events Crew

September 2025-Present

- Operate light and sound boards for Cornell's PMA events and showcases

Campus Involvement

- Volunteer Sitter - Guiding Eyes for the Blind *September 2024-Present*
- Full Technical Director - Cog Dog Theater Troupe *August 2025-Present*
- Safety Officer - Design Build Fly *April 2026-Present*

Specialized Skills

Solidworks, Fusion 360, Onshape, Ansys, Vectorworks, Procore, Microsoft Office Utilities, Basic Python and MatLab, Red Apron Trained- Emerson Manufacturing Lab, machine shop tools and hand power-tools